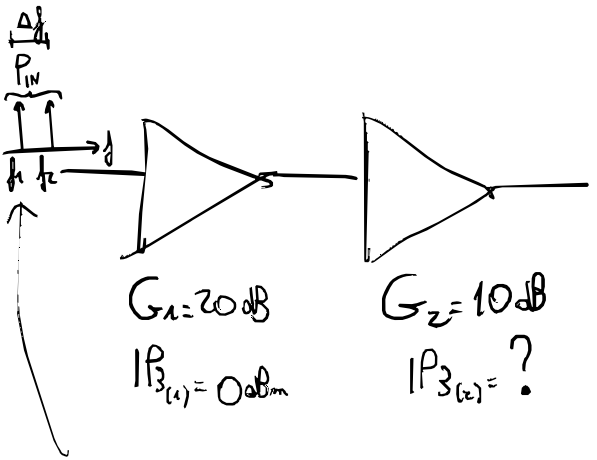


CI = 40 dB

(USE PURE 3rd ORDER NON LIN. MODEL)

INPUT SIGNAL:
a 2-TONE SIGNAL with AVERAGE POWER $P_{in} = -40$ dBm
CENTERED AT $f = 2.4$ GHz, with SPACING 40 MHz = Δf

(1) BACK OFF at which the FIRST AMPLIFIER is WORKING?

(2) DRAW SPECTRUM @ OUTPUT of AMPLI 1 (frequencies & POWERS)

(3) DRAW SPECTRUM @ OUTPUT of AMPLI 2

(4) WHAT $IP_{3(2)}$ SHOULD HAVE AMPLI 2
to guarantee that $C/I = 40$ dB? (NEGLECT COMPRESSION of MAIN TONES)

(1) BACKOFF at which the FIRST AMPLIFIER is WORKING?

BO = "HOW FAR is the OUTPUT POWER of DESIRED SIGNAL from P_{1dB}"

ASSUMING we can neglect COMPRESSION of MAIN TONES:

P_{m_1} = OUTPUT POWER of DESIRED SIGNAL AFTER AMPLI 1

$= P_{IN} + G_1 = -20\ dB_m$

$P_{1dB_{(1)}} = IP_{3(1)} - \Delta_p = -10.63\ dB_m$

↖ 10,63dB for PURE 3rd ORDER MODEL

$BO_{(1)} = P_{1dB_{(1)}} - P_{m_1} = 9.4\ dB$

(⇒ LARGE BACKOFF, we can really neglect COMPRESSION of MAIN TONES)

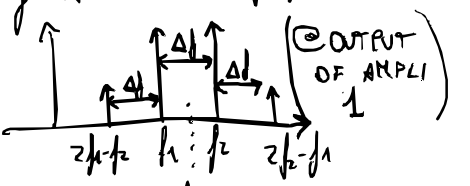
(2) DRAW SPECTRUM @ OUTPUT of AMPLI 1 (frequencies & POWERS)

$f_1 = f_0 - \frac{\Delta f}{2} = 2.38\,GHz$

$f_2 = f_0 + \frac{\Delta f}{2} = 2.42\,GHz$

$2f_1 - f_2 = 2.34\,GHz = f_0 - \frac{3}{2}\Delta f$

$2f_2 - f_1 = 2.46\,GHz = f_0 + \frac{3}{2}\Delta f$

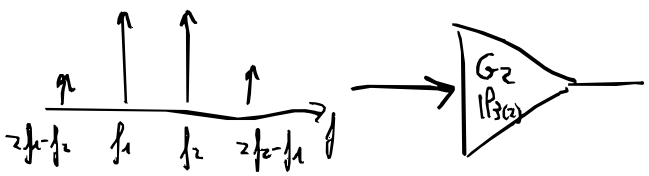


$P_{w_1} = P_{w_2} = P_{in1} - 3dB = -23\,dBm$

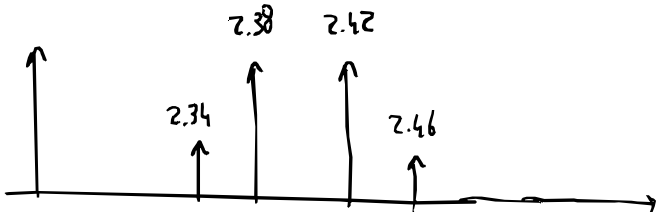
$P_{2w_1-w_2} = P_{2w_2-w_1} = 3P_{w_1} - 2IP_{3(1)} = -69\,dBm$

(dBm)

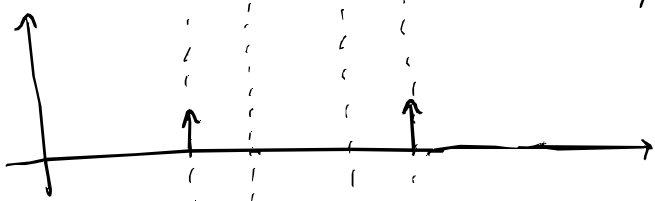
③ DRAW SPECTRUM @ OUTPUT of AMPLI 2
AND WRITE EXPRESSION of $P_{2f_1-f_2}$ AT ITS OUTPUT



LINEAR AMPLIFICATION BY G_2



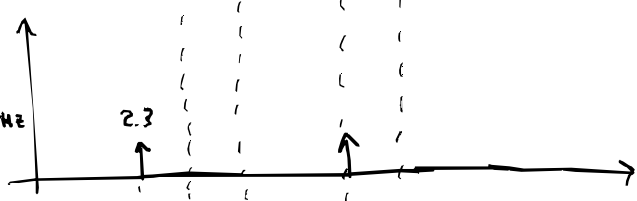
INTERMOD of f_1, f_2



INTERMOD of $2f_1-f_2, f_1$

→ $2(2f_1-f_2)-f_1 = 3f_1-2f_2 = 2.3\,GHz$

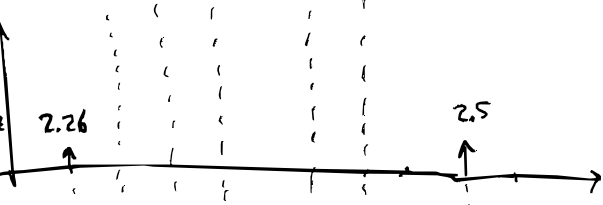
→ $2(f_1)-(2f_1-f_2) = f_2$



INTERMOD of $2f_1-f_2, f_2$

→ $2(2f_1-f_2)-f_2 = 4f_1-3f_2 = 2.26\,GHz$

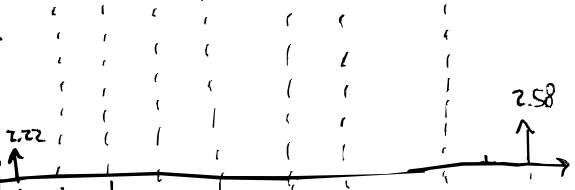
→ $2f_2-(2f_1-f_2) = 3f_2-2f_1 = 2.5\,GHz$



INTERMOD of $2f_1-f_2, 2f_2-f_1$

→ $2(2f_1-f_2)-(2f_2-f_1) = 5f_1-4f_2 = 2.22\,GHz$

→ $2(2f_2-f_1)-(2f_1-f_2) = 5f_2-4f_1 = 2.58\,GHz$



INTERMOD $f_1, 2f_2-f_1$

→ $2f_1-(2f_2-f_1) = 3f_1-2f_2 = 2.3\,GHz$

→ $2(2f_2-f_1)-f_1 = 4f_2-3f_1 = 2.54\,GHz$



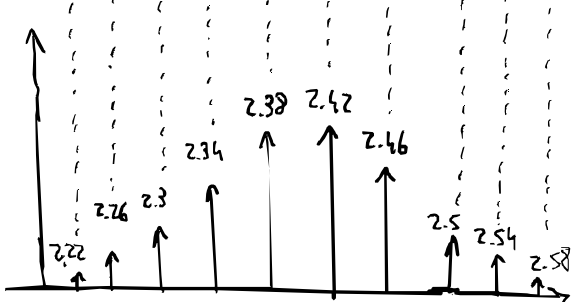
INTERMOD $f_2, 2f_2-f_1$

→ $2f_2-(2f_2-f_1) = f_1$

→ $2(2f_2-f_1)-f_2 = 3f_2-2f_1 = 2.5\,GHz$



TOTAL SPECTRUM QUALITATIVELY



(4) WHAT $IP_{3(2)}$ SHOULD HAVE AMPLI 2 to guarantee that $C/I = 40\,dB$? (NEGLECT COMPRESSION of MAIN TONES)

We can compute the required $IP_{3\,TOT}$ to guarantee $C/I = 40\,dB$ AT THE OUTPUT. Then we can use the FORMULA for CASCADE to FIND $IP_{3(2)}$

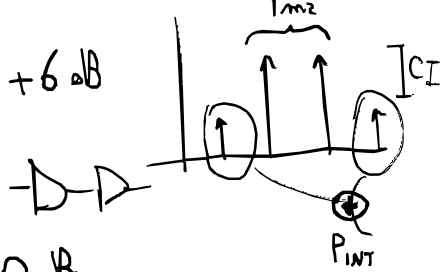
$$CI_{dB} = \text{"POWER of USEFUL SIGNAL @ OUTPUT"}_{(dBm)} - \text{"POWER of UNDESIRED INTERMOD"}_{(dBm)}$$

$$= P_{m_2}\,(dBm) - P_{INT}\,(dBm) = P_{m_2} - \left(P_{2w_2-w_1} + 3\,dB\right)$$

$$= P_{m_2} - \left(3\,P_{w1\ @\ OUTPUT} - 2\,IP_{3_{TOT}} + 3\,dB\right)$$

$$= P_{m_2} - \left(3\,P_{m_2} - 2\,IP_{3\,TOT} - 9\,dB + 3\,dB\right)$$

$$= 2\,IP_{3\,TOT} - 2\,P_{m_2} + 6\,dB$$



$P_{m_2} = -40\,dBm + G_1 + G_2 = -10\,dBm$

$$\Rightarrow IP_{3\,TOT} = \frac{CI_{dB} - 6\,dB + 2P_{m_2}}{2} = 7\,dBm = 5\,mW$$

(THEN WE COMPUTE $IP_{3(2)}$) ⟹

MORE PRACTICAL FORMULA

$$\left(\frac{1}{iP_{3TOT}}\right)^2 = \left(\frac{1}{g_2 iP_{3(1)}}\right)^2 + \left(\frac{1}{iP_{3(2)}}\right)^2$$

$$iP_{3(2)} = \frac{1}{\sqrt{\left(\frac{1}{iP_{3TOT}}\right)^2 - \left(\frac{1}{g_2 iP_{3(1)}}\right)^2}} = 5.77\ mW = 7.6\ dBm$$

VERIFY CI

THE MAIN TONES ARE EACH:

$P_{w1} = P_{w2} = -23\ dBm + 10\ dB = -13\ dBm$
$@OUT_2 \quad @OUT_2$

THE DOMINANT INTERMOD TONES REMAIN THOSE AT $2f_1 - f_2$ AND $2f_2 - f_1$ THUS ARE:

— LINEAR AMPLIFICATION of INTERMOD TONES from AMPLI 1
⇒ $g_2 \cdot P_{2w_2-w_1}$ (IN LINEAR SCALE): $-69\ dBm + 10\ dB = -59\ dBm = 1.26\ mW$
$@OUT_1$

— + FURTHER INTERMODULATION by f_1, f_2
⇒ $\left(3 P_{out w_1} - 2IP_{3(2)}\right)$ → CONVERTED TO LINEAR SCALE: $-54\ dBm = 4\ mW$
$@OUT_2$

⇒ $P_{2w_2-w_1} = 1.26\ mW + 4\ mW = 5.26\ mW$
$@OUT\ 2$
$= -52.8\ dBm$

[ASSUMING INTERMOD. TONES ADD IN POWER, PRACTICAL ASSUMPTION]

⇒ $CI = -13\ dBm - (-52.8\ dBm) \simeq 40\ dB$: OK!

WORST CASE FORMULA, MORE CONSERVATIVE

$$\frac{1}{iP_{3TOT}} = \frac{1}{g_2 iP_{3(1)}} + \frac{1}{iP_{3(2)}} \Rightarrow iP_{3(2)} = \frac{1}{\frac{1}{iP_{3TOT}} - \frac{1}{g_2 iP_{3(1)}}} = 10\ mW$$

$IP_{3(2)} = 10\ dBm$ (HIGHER REQUIREMENT ON AMPLI 2 THAN WITH MORE PRACTICAL FORMULA)